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Research Highlights

The relationship between thermal management methods and hydrogen storage performance of the metal hydride tank

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ABSTRACT

Solid-state hydrogen storage tanks are key equipment for fuel cell vehicles and hydrogen storage. However, the low heat transfer properties of hydrogen storage tanks result in the inability to meet the hydrogen supply requirements of fuel cells. In this study, different thermal management approaches were explored through the design of $LaNi_5$-based solid-state hydrogen storage tanks. We experimentally studied the effects of different internal heat transfer methods, that is, expanded natural graphite (ENG), copper foam, and copper fins on the hydrogen absorption and desorption performance. We also studied the effects of external cooling methods with natural convection, air cooling, and water cooling, respectively. Under the same external cooling method of natural convection, a solid hydrogen storage tank filled with 5 wt.% ENG has similar performance to a tank filled with copper foam. Compared to natural convection, air and water cooling can significantly improve the heating performance of metal hydride (MH) beds by increasing the external heat transfer coefficient. The effect of water cooling is better than that of air cooling, and in these two enhanced performance conditions, the tank filled with copper foam performs better than with ENG. In the case of water cooling, by adding copper fins to a hydrogen storage tank filled with 5 wt.% ENG, the tank was saturated with hydrogen absorption in only 29.4 min, which is 55.6 % shorter than the hydrogen uptake time in a hydrogen storage reactor without copper fins. And its stable hydrogen desorption (1 NL/min) has reached 98.1 % of the total hydrogen released. The results show that the effective thermal conductivity and heat transfer area of metal hydride bed play key roles in improving heat transfer and reaction rate. In addition, heat transfer is more important than mass transfer to improve the performance of the hydrogen storage tank.

1. Introduction

Sustainable, clean, and efficient energy needs to be emphasized in the face of global warming [1,2]. Common energy storage technologies, such as hydrogen storage, bioenergy storage, and electrochemical energy storage, return waste heat from industry, wind, and sunshine into renewable energy sources [3–6]. In this situation, hydrogen energy is an important bridge connecting the transition from fossil energy to renewable energy [7,8], and it is an important green energy carrier in the 21st century [9,10]. Hydrogen has many uses, for example, it can be used for energy storage, directly used by fuel cell to produce electricity, or burned to provide energy [11–14]. However, hydrogen storage is one of the biggest challenges for global hydrogen energy [15,16]. Conventional hydrogen storage techniques, like gaseous and liquid hydrogen storage systems, cannot meet the requirements of the hydrogen economy in the future because of their safety and cost concerns [17]. The solid-state hydrogen storage methods have attracted a lot of interest [18–21] due to their high volumetric hydrogen density, stability, low cost, and safety [22–26].

The hydrogen absorption/desorption performance of the metal hydride (MH) tank is very important for the development of solid-state hydrogen storage technology. The hydrogen

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absorption/desorption process in the MH tanks can be divided into the hydrogen absorption/desorption process of the alloy and the heat and mass transfer process in MH tanks [27–29]. The hydrogen absorption and desorption performances of alloys are mainly determined by their thermodynamics [21,30–33] and kinetic properties [34–38]. The hydrogen storage properties of the alloys have been enhanced mainly through elemental substitution and different preparation methods [39–43]. $LaNi_5$-based alloys are among the most promising metal hydrides because of their quick kinetics, ease of activation, and ability to absorb and desorb hydrogen at room temperature [44,45]. However, the hydrogen absorption/desorption rate of MH tanks mainly depends on the heat and mass transfer process in MH tanks because of the tanks' inadequate heat and mass transfer properties [46–48]. Therefore, the mass and heat transfer performances of MH tanks need to be improved.

According to Fourier's law, increasing the effective thermal conductivity (by metal foam, expanded natural graphite), heat transfer areas (by fins), and heat transfer coefficients (by cooling tubes, external cooling method) of MH tanks can improve the heat transfer behavior of hydrogen storage tanks [27,29,49–55]. Metal foam (MF) contains foam pores, which are filled with hydrogen storage alloys to improve the effective thermo-physical properties of the MH bed [50,51]. Tsai et al. [56] numerically studied the hydrogenation performance of MH tanks with aluminum foam having different porosities at the same mass of hydrogen storage alloy powder. The results showed that Al foam at 1 % volume fraction was significant in improving the hydrogen absorption properties. Bai et al. [57] numerically investigated tanks with gradient porosity MF and uniform MF, respectively. The results showed that the hydrogen absorption properties of the MH tank with gradient porosity foam metal were improved by 11.5 %. The addition of expanded natural graphite (ENG) can improve the thermal conductivity and the mechanical stability of metal hydride beds [52,53]. The metal hydride-ENG composites developed by Dieterich et al. [58] maintained high mechanical integrity after 1000 cycles. Singh and Bhogilla [59] studied a MH hydrogen storage tank with a diameter of 35 mm and a height of 450 mm. The hydrogen absorption saturation time of the tank with ENG was 51 % shorter than that of the tank without ENG. Klein and Groll [60] discussed the improving performance of metal hydride-ENG composites and the effect of aluminum foam on the hydrogen charge and discharge of MH tanks. The simulation results showed that the metal hydride-ENG composites outperform the aluminum foam when the mass ratio of MH to ENG is > 20 % and the porosity of the aluminum foam is about 70 %. The addition of fins will increase the heat transfer area of the metal hydride bed [54,55,61,62], and the thermal reinforcement of fins is better than that of metal foam [54]. Zhang and co-workers [55] discussed the optimization of heat transfer enhancement in MH tank with straight tube heat exchangers and longitudinal fins by numerical methods. The results showed that the increase of longitudinal fins can effectively enhance the overall heat transfer and reduce the reaction time by 25 %. Moreover, studies have focused on the combination of heat transfer tubes and fins to improve heat transfer [63–67].

The main research of heat and mass transport issues in hydrogen storage tanks nowadays are predicted by simulations, where certain tanks with good performance are hard to implement in practicality. Thus, in this work, three easily filled tanks were designed to experimentally discuss the effects of different internal cooling methods (ENG, copper foam, and ENG & copper fins) and external cooling methods (natural convection, air cooling, and water cooling) on the heat transfer performance of MH hydrogen storage tanks. Furthermore, in contrast to the traditional hydrogen desorption under constant pressure, the hydrogen desorption at the constant flow described in this research simulates the effective hydrogen release from the hydrogen storage tank when it works with the fuel cell. These experimental results can have strong implications for the improvement of hydrogen storage performance in the future design of MH tanks.

2. Experimental

2.1. Tank design

The schematic diagram and photograph of the MH tank are shown in Fig. 1. The tank was made of 304 stainless steels with an inner diameter and an inner height of 70 mm and 130 mm, respectively, and a wall thickness of 5 mm. Due to the volume expansion of hydrogen storage alloy after hydrogen absorption [68], the actual filled height was 100 mm. The sealing of the tank was ensured by the flange compression O-ring. The hydrogen passed into and out of the tank through a porous sintered filter with a di-

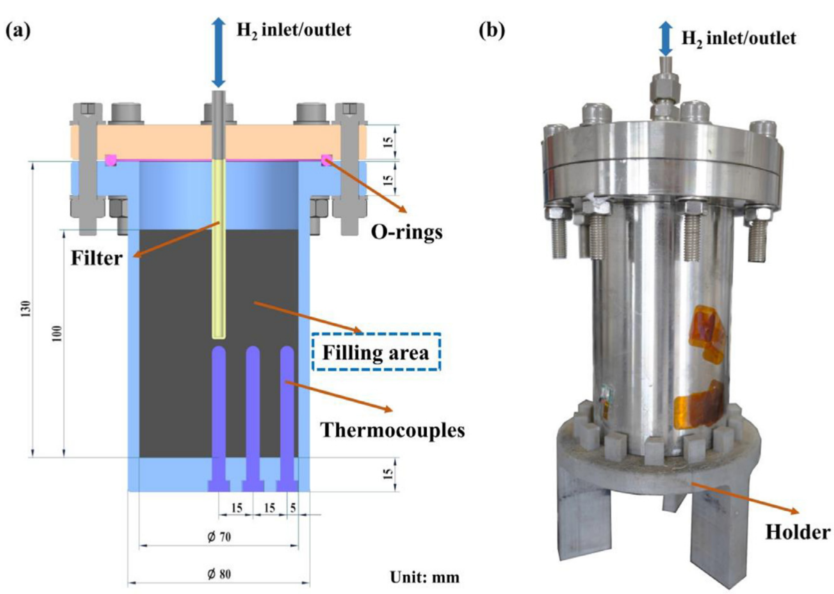

Fig. 1. (a) Schematic diagram and (b) photograph of MH hydrogen storage tank.

Table 1
Axial and radial locations of thermocouples.

Thermocouple	Radial location from center (mm)	Axial location from base (mm)
TC1	0	50
TC2	15	50
TC3	30	50
TC4	40	50
TC5 (Environment)	\	\

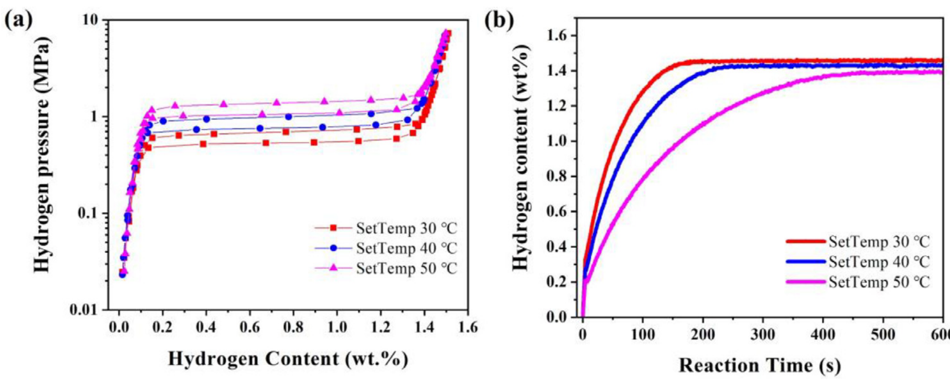

Fig. 2. (a) Pressure composition isotherm and (b) reaction kinetics for La-Ce-Ni-Co-Mn.

ameter of 6 mm and a pore size of 1–3 μm to prevent MH alloy from entering the hydrogen pipeline. 5 K-type thermocouples were used to detect the change of the temperature field during hydrogen absorption and desorption from the MH tank. The place of these thermocouples is listed in Table 1. Three thermocouples were used to monitor the MH bed. The remaining thermocouples were used to detect the tank wall and the environment.

Hydrogen was stored in 1150 g of commercially available La-Ce-Ni-Co-Mn alloy ($LaNi_5$-based alloy) supplied by Xiamen Tungsten Co., Ltd., China. Pressure-composition-temperature (PCT) curves and absorption/desorption kinetics were measured on a fully automated Sieverts-type apparatus (H2PCT-2151, GASTOUCH) [69,70]. Fig. 2 shows the PCT and reaction kinetic curves for the La-Ce-Ni-Co-Mn at 30, 40, and 50 °C. The platform pressure and hydrogen content of the MH alloy are 0.69 MPa and 1.45 wt.%, respectively, at 30 °C in Fig. 2(a). And the alloy can absorb hydrogen within 200 s at 30 °C under 2.0 MPa H_2 as shown in Fig. 2(b).

In this work, the influences of three types of filling structures on the heat and mass transfer in the MH tank were investigated. As shown in Fig. 3, three designs are considered:

- Case 1: an MH tank equipped with 5 wt.% expanded graphite (Fig. 3(a)).
- Case 2: an MH tank equipped with copper foam (Fig. 3(b)).
- Case 3: based on Case 1, assembling of five copper fins with a thickness of 0.5 mm (Fig. 3(c)).

A blank control group Case 0 (only filled 1150 g La-Ce-Ni-Co-Mn alloy without internal thermal management) was designed. In addition, for each of the four MH tanks, we also studied the effects of their external cooling methods such as natural convection, air cooling, and water cooling, respectively.

2.2. Experimental apparatus

A testing system of the properties of the MH tank during the charging and discharging process was built. The hydrogen source of the system consists of a 40 L high-pressure cylinder and a pressure-reducing valve. The pipeline and the MH tank were evacuated with a vacuum pump before absorbing the hydrogen. A pressure transducer close to the MH tank was used to monitor the pressure. A mass flow controller (M040-LK2, Alicat) with a measuring range of 0–20 NL/min and an accuracy of 0.8 % of reading +0.2 % of full scale was used to control the rate at which the hydrogen is discharged. Due to the maximum pressure resistance of the mass flow controller of 1 MPa, a pressure-reducing valve is installed between the mass flow controller and the MH tank to protect the mass flow controller. Signal ports for pressure transducer, thermocouples, and mass flow controller are connected to a computer for data collection.

Fig. 4 shows the schematic of the hydrogen storage system. The hydrogen absorption testing procedure is described as follows. Firstly, the pipeline and the tank were evacuated by a vacuum pump for 6 h through V2, V7, and V8. Secondly, turn on the thermocouples and pressure transducer collection data system. Finally, hydrogen at 2.0 MPa was fed into the tank through valves V1 and V2 until full saturation with hydrogen, during which time the other valves remained closed. The hydrogen desorption testing procedure is described as follows. The mass flow controller was set to control the flow rate at 1 NL/min and the data (temperature, pressure, and flow rate) acquisition system was switched on. It opened valves V2, V3, V4, and V5, allowing the hydrogen to be directly emitted into the atmosphere.

3. Results and discussion

3.1. Hydrogen absorption process of MH tanks

In this work, the hydrogen absorption process of the MH tanks is a constant pressure absorption, and according to Fig. 2, the hydrogen supply pressure of the MH tanks is set to 2.0 MPa. The hydrogen absorption reaction in solid hydrogen storage alloys is exothermic. It is possible to watch the hydrogen absorption and desorption process in the tank by monitoring the temperature of the tank [71].

3.1.1. Effect of radial location

For alloy property testing, the powder bed scale is small and the differences in temperature and pressure at different locations within the bed can be ignored [72]. However, in an MH tank, due to the low effective thermal conductivity of the powder bed [73],

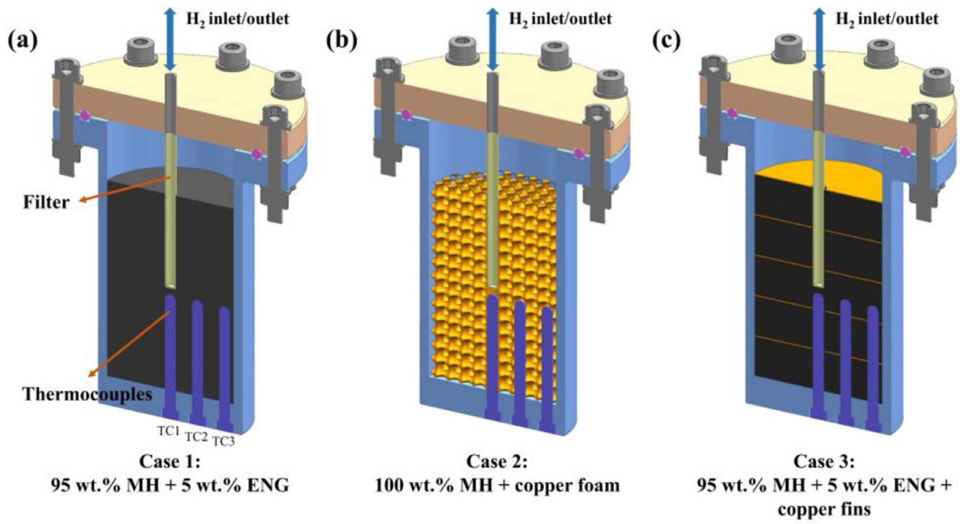

Fig. 3. Three MH tanks with (a) Case 1: 95 wt.% MH + 5 wt.% ENG, (b) Case 2: 100 wt.% MH + copper foam, and (c) Case 3: 95 wt.% MH + 5 wt.% ENG + copper fins.

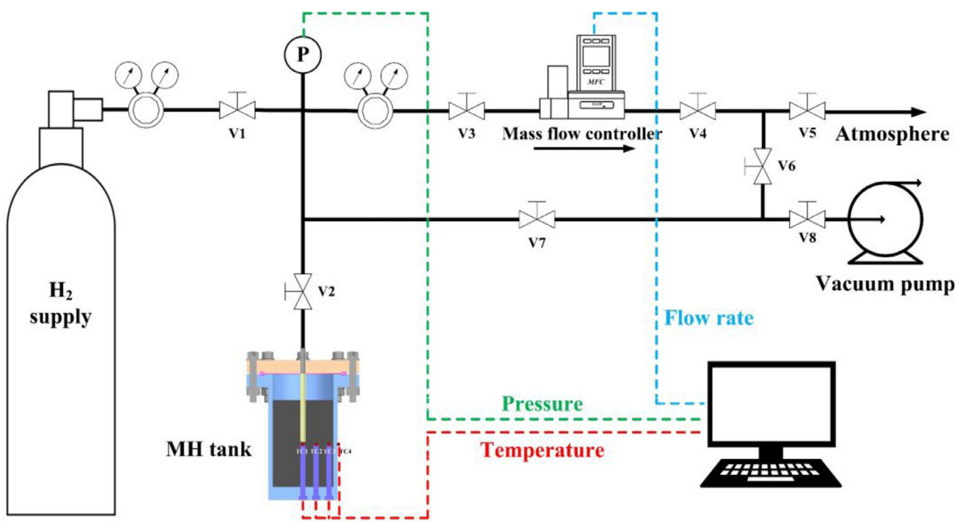

Fig. 4. Schematic diagram of the hydrogen storage system.

the heat generated from the hydrogen absorption reaction could not be discharged in time, and the temperature in the internal region of the powder bed increased dramatically, which in turn led to an obvious temperature gradient in the powder bed. Fig. 5 shows the evolution of different radial temperatures for the hydrogen absorption process. The closer to the center of the MH beds, the higher the temperature and the lower temperature at the tank wall. It was also found in the study by Laurencelle and Goyette [50] and Singh et al. [62]. The difference between these can be attributed to the fact that it is more difficult to transfer heat efficiently from the center of the MH tank than from the walls.

From Fig. 5, we also find that the temperature near the wall of the tank decreases earlier than near the center of the tank. This phenomenon was also developed by Ye et al. [74] in their simulation data. This is due to the fact that the alloy in the tank is constantly exothermic during hydrogen absorption, and the heat produced from within needs to be diffused outwards in order to cool the alloy for continued hydrogen absorption. The absorption of hydrogen is known to increase with decreasing temperature at the same absorption pressure [75]. So, the completion of hydrogen absorption needs to take place when the temperature of the alloy is lowered to the stabilization temperature. And the alloy releases heat when absorbing hydrogen. When the heat cannot be conducted quickly enough, it causes the temperature of the alloy to increase. Whereas for MH hydrogen storage tanks, heat is a process that is exported from the inside to the outside. When the heat can't be quickly and efficiently exported, it will be enriched inside of the tank causing the internal temperature of the tank to rise. The hydrogen absorption platform pressure of the alloy, in turn, increases with temperature [76]. It leads to the fact that at a certain hydrogen supply pressure, when the alloy rises to a certain temperature, it stops absorbing hydrogen and continues to do that only when the temperature is reduced. Therefore, the external alloy is easier and preferred to be lowered to the stabilization temperature, i.e., the lowest temperature. And the center of the tank is closest to the filter (hydrogen transmission channel). Based on the above discussion, it can be conclued that heat transfer is more important than mass transfer to improve the performance of the hydrogen storage tank, which also was supported by the reported simulation data [61,65,77].

In addition, the hydrogen absorption process of the alloy can be summarized in 4 stages:

Stage I: Hydrogen storage alloys react rapidly, releasing large amounts of heat during the initial phase of contact with hydrogen. The temperature rises rapidly to the highest temper-

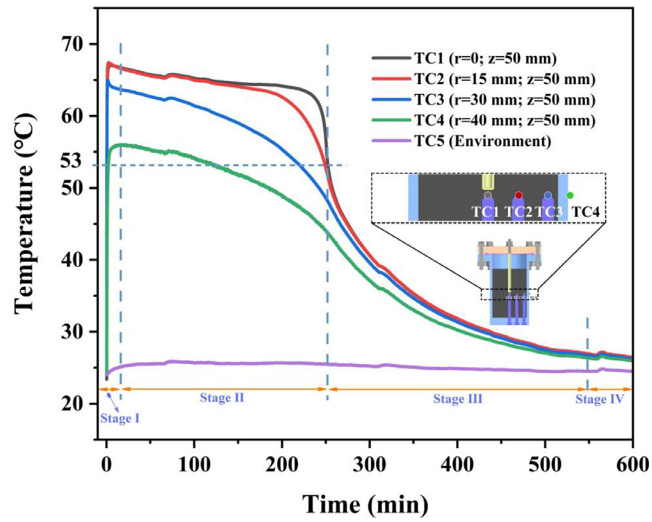

Fig. 5. Evolution of different radial temperatures for hydrogen absorption processes.

ature, and the center of the tank temperature (TC1 and TC2) could reach 67 °C. Temperatures near the tank wall (TC3) and at the tank wall (TC4) were 65 and 56 °C, respectively. TC3 has a lower temperature than TC1 and TC2 because of its location closer to the tank wall, where the heat can be transferred quickly. The highest temperature of TC1 and TC2 rise suggested that the heat in the center part is difficult to be transferred outwards. According to the Van't Hoff equation, the equilibrium pressure equation is [29,55,78],

$$P_{\mathrm{eq}} = f(H/M)\exp\left[-\frac{\Delta H}{R_{\mathrm{g}}}\left(\frac{1}{T}-\frac{1}{T_{\mathrm{ref}}}\right)\right] \tag{1}$$

where P_{eq} is the platform pressure, $f(H/M)$ is the polynomial fitting equation of the PCT at the reference temperature (T_{ref}), H/M is the hydrogen content, and ΔH is the enthalpy change. Fig. 6 shows the predicted pressure composition isotherms of hydrogen absorption by Eq. (1). Just 0.145 wt.% H_2 can be absorbed under 2.0 MPa H_2 at 67 °C and the equilibrium plateau pressure at 67 °C is higher than 2.0 MPa, which is the reason that the highest temperature

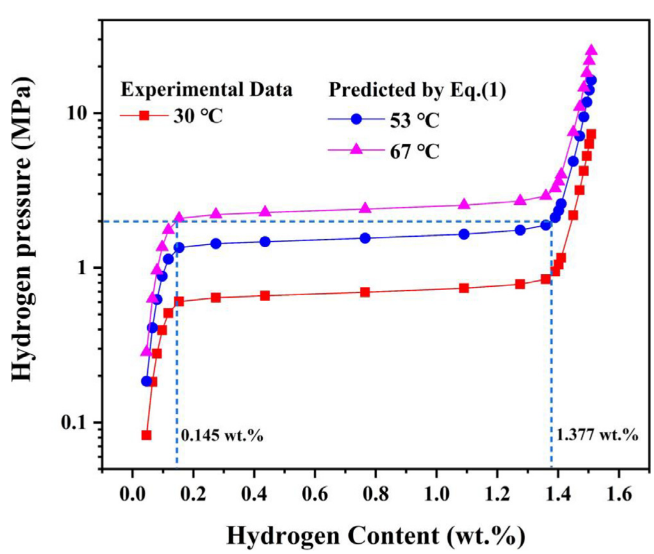

Fig. 6. Predicted pressure composition isotherms at hydrogen absorption.

during the absorption process in the MH tank under 2.0 MPa H_2 is 67 °C.

Stage II: This stage is the main stage of hydrogen absorption. The temperatures of TC1 and TC2 were essentially the same, maintained above 60 °C, and the temperatures of TC3 and TC4 decreased rapidly. As hydrogen absorption continued, TC1 and TC2 stabilized up to 220.1 and 184.4 min, respectively, and then began to decline rapidly. At 252.5 min, the temperature of TC1 was reduced to 53 °C. The temperature of TC2 converged to TC1 at this time due to lower heat transfer from the core of the tank. The hydrogen adsorption of 1.377 wt.% at a supply pressure of 2.0 MPa at 53 °C is shown in Fig. 6, which corresponds to 95 % of the maximum hydrogen storage capacity under 20 MPa H_2. Therefore the hydrogen storage capacity of the tanks is higher than 95 %. Combining Figs. 5 and 6, it can be seen that the alloy close to the tank wall (the location of TC3) has aborbed more than 95 % hydrogen within this stage, while alloy close to the core (the locations of TC1 and TC2) has just absorbed 95 % hydrogen. This is because that the alloy near the wall of the tank at this stage is cooling down faster and absorbing hydrogen at a faster rate due to faster heat transfer. The heat transfer performance of the tank is far from enough to meet the heat release rate in the process of hydrogen absorption of the alloy close to the core of the tank. Therefore, at the beginning of this stage, the temperature near the core changed little, corresponding to a small hydrogen absortpion. With the temperature near the wall decreasing below 53 °C, corresponding to an almost completed hydrogen absorption, the temperature near the core started to quickly decreased to 53 °C, indicating that the alloy near the core absorbed 95 % H_2 finally at the end of this stage. Therefore, Stage II is mainly a process in which the hydrogen absorption of the alloy gradually reaches more than 95 % from the wall to the core of the tank.

Stage III: At the end of stage II, the alloy was saturated with hydrogen absorption, no more heat was released and the temperature decreased. However, according to the Van't Hoff equation, the lower temperature causes the P_{eq} to become smaller, leading to higher hydrogen storage capacity [75]. The alloy continues to absorb hydrogen, but the amount of hydrogen adsorbed is not large enough to keep the heat released more than the heat dissipated from the tank. That's why TC1-4 started to fall steadily and the temperature difference between each other is getting smaller and smaller.

Stage IV: The hydrogen absorption reaction in the tank was complete and the temperature returned to room temperature.

The above analysis shows that at the end of stage II, the hydrogen storage tank can achieve more than 95 % of the hydrogen storage capacity. As a result, in order to enhance the properties of the tank, the time of stage II should be decreased as much as possible. The end time of stage II can be considered as the end of hydrogen absorption.

3.1.2. Effect of internal heat transfer methods

Fig. 7 shows the effects of different internal heat transfer methods on the temperature of the hydrogen absorption process. The supply pressure was fixed at 2.0 MPa for all tanks with different internal configurations under different external environments. The hydrogen absorption time with different thermal management modes is shown in Fig. 8, where the time point is measured at the end of stage II.

As shown in Fig. 8, the hydrogen absorption times for Case 1 (252.2 min) and Case 2 (251.8 min) under natural convection

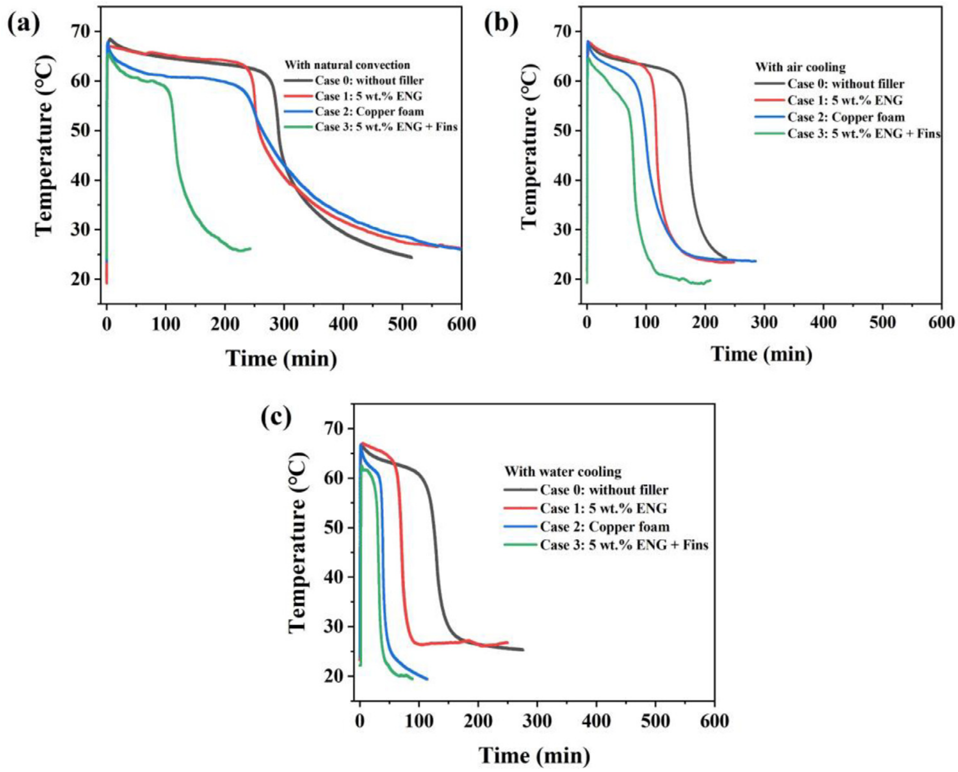

Fig. 7. The effects of different internal structures on the temperature of the hydrogen absorption process: (a) with natural convection; (b) with air cooling; (c) with water cooling.

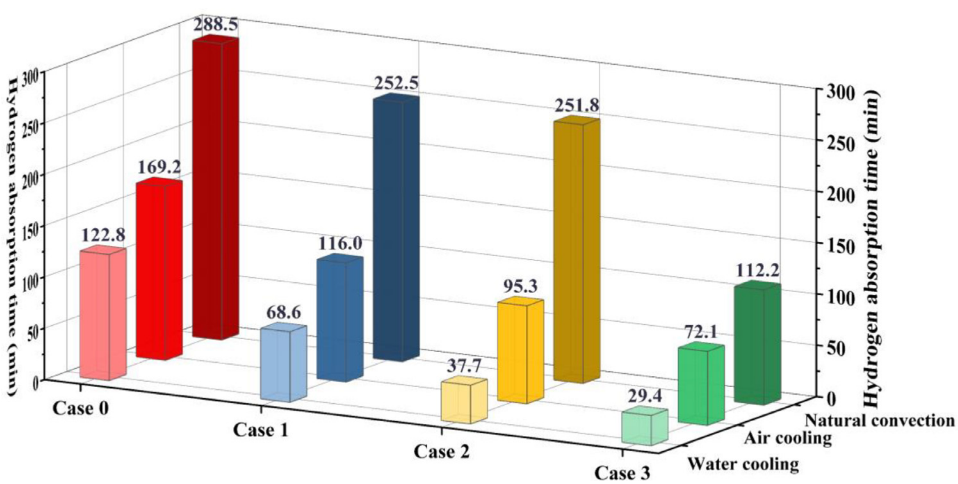

Fig. 8. The hydrogen absorption time of MH tanks with different thermal management methods.

condition are similar, which are 13 % shorter than that of Case 0 (288.5 min) without internal heat transfer structure. These results indicate that the effective thermal conductivity of the MH tanks can be effectively increased by the addition of ENG or copper foam. It is noted that the hydrogen absorption times for Case 1 and Case 2 under natural convection condition are similar although the heat conductivity of Cu is higher than that of ENG. This indicates that under the natural convection condition, the hydrogen absorption rate controlling step for the MH tank is the outside heat transfer of the tank. By adding copper fins to Case 1, the hydrogen absorption time of Case 3 is reduced by 55.6 % compared with that of Case 1. This is because the addition of copper fins increases the heat transfer area. From Figs. 7, 8, it can be seen that the maximum temperatures and the hydrogen absorption times of the MH tanks decrease in the order of Case 0 > Case 1 > Case 2 > Case 3 under air cooling and water cooling conditions. Normally, the better the heat transfer, the lower the temperature, and thus the faster the hydrogen absorption rate of the MH tank. These results showed that the order of heat transfer performance of the MH tanks with different internal heat transfer structures is Case 0 < Case 1 < Case 2 < Case 3.

It is clear that the best overall heat transfer performance was obtained for Case 3, which was saturated with hydrogen absorption in only 29.4 min with water cooling. The improvement of hydrogen absorption time for Case 3 is 55.4 %, 24.3 %, and 22.0 % with natural convection, air cooling, and water cooling, respectively, even compared with Case 2 filled with copper foam. These results demonstrate how important the internal heat trans-

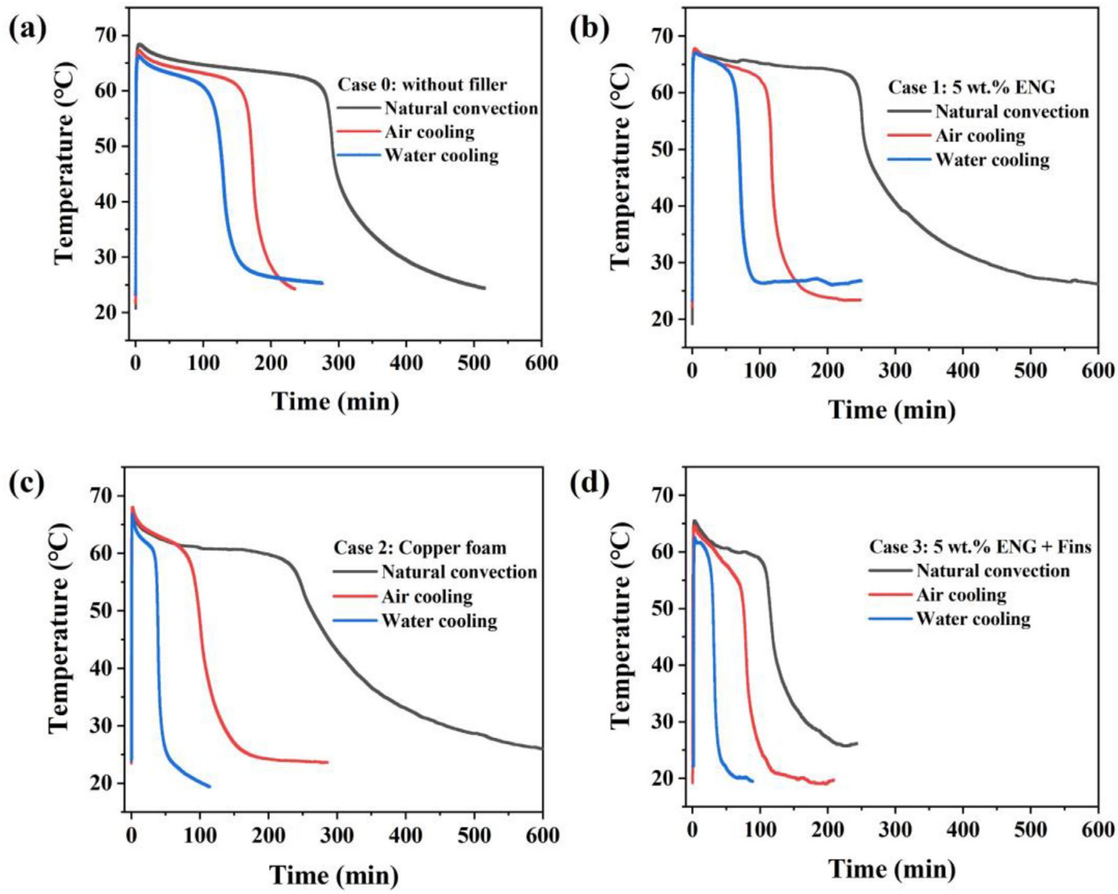

Fig. 9. The effects of different heat transfer methods on the temperature of the hydrogen absorption process: (a) Case 0; (b) Case 1; (c) Case 2; (d) Case 3.

fer methods are for improving the hydrogen absorption performance of the hydrogen storage tank.

3.1.3. Effect of external heat transfer methods

Fig. 9 shows the effects of different external heat transfer methods on the temperature of the hydrogen absorption process. The hydrogen absorption time with different external heat transfer methods is shown in Fig. 8, where the time point is measured at the end of stage II.

Obviously, it can be found that the hydrogen absorption time of the MH tanks decreases in the order of natural convection > air cooling > water cooling for all four kinds of internal heat transfer methods. This is mainly because the heat transfer coefficient of the external heat transfer methods increases in the order of natural convection < air cooling < water cooling. When the external cooling method changes from natural convection to air-cooling, the improvement of hydrogen absorption time for Case 0, Case 1, Case 2, and Case 3 is 41.4 %, 54.1 %, 62.2 %, and 35.7 %, respectively. The improvement of hydrogen absorption time becomes 57.4 %, 72.8 %, 85.0 %, and 73.8 %, respectively, when the air-cooling changed into water cooling. According to Fourier's Law and Newton's Law,

$$q = kA\frac{dT}{dx} = hA\Delta T \tag{2}$$

where the convective heat transfer coefficient h is directly proportional to the convective heat transfer rate q, k is the effective thermal conductivity, and A is the heat transfer area. The heat transfer coefficients for natural convection, air cooling, and water cooling are 5–25, 20–300, and 200–1000 W/(m^2 K), respectively [61,79]. A higher heat transfer coefficient of water cooling would lead to a faster heat transfer rate at the tank wall, which could accelerate the hydrogen absorption rate.

It is worth noting that the improvement of hydrogen absorption properties of the tank with higher effective thermal conductivity of the MH beds is even more obvious when the external heat transfer coefficient increased in Fig. 8. Compared to the blank control group (Case 0), when the external heat transfer methods from natural convection to water cooling, the hydrogen absorption time for Case 1 was decreased by 12.5 %, 31.4 %, and 44.1 %, and the hydrogen absorption time for Case 2 was decreased 12.7 %, 43.7 %, and 69.3 %. The improvements in performance are more pronounced for Case 2 than for Case 1 under the air cooling and water cooling, because the thermal conductivity of the MH beds increases in the order of Case 0 < Case 1 < Case 2. These results indicate that when the external heat transfer coefficient is improved, the hydrogen absorption rate controlling step for the MH tank changes to the internal heat transfer from the external heat transfer.

3.2. Hydrogen desorption process of MH tanks

In this work, considering the hydrogen storage tanks coupled with a fuel cell application [46,71], a mass flow controller is used to control the hydrogen discharge flow rate to 1 NL/min to simulate the fuel cell operating condition. The pressure at the hydrogen outlet is 0.1 MPa. Hydrogen desorption is a heat absorption process. In this section, the hydride bed reaction state is monitored by temperature, flow, and pressure data.

3.2.1. Effect of radial location

Fig. 10 shows the evolution of flow rate, pressure, and different radial temperatures during hydrogen desorption for Case 1. The hy-

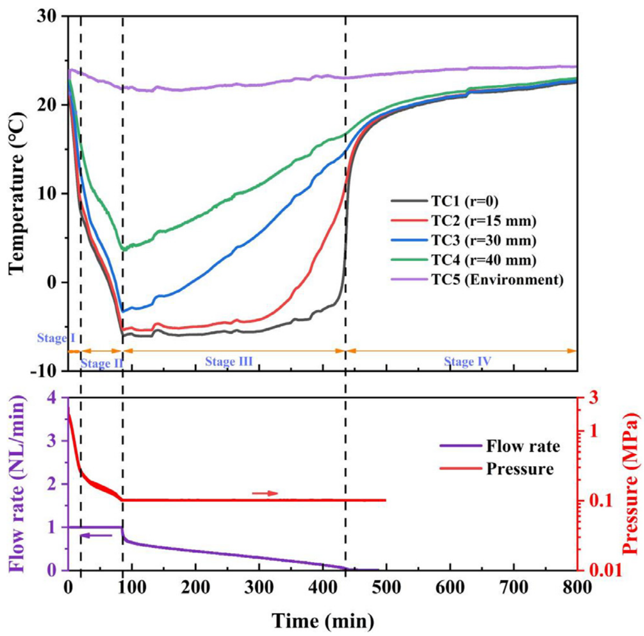

Fig. 10. Evolution of flow rate, pressure, and different radial temperatures for hydrogen desorption processes.

drogen release reaction is a heat absorption reaction, and the phenomenon is the opposite of the hydrogen absorption reaction, with lower temperatures near the center of the tank and higher temperatures at the walls of a tank with good heat transfer properties.

The constant flow hydrogen desorption process can also be organized into four stages:

Stage I: Initial stage of hydrogen release. The temperature and pressure drop rapidly, and unlike hydrogen absorption, the temperature does not drop to its lowest point. This is consistent with the results of Chung et al. [71]. The reason is that this stage is the process of reducing the pressure in the hydrogen storage tank to the P_{eq}, i.e., the transformation of β-phase to ($\alpha + \beta$)-phase [75].

Stage II: Stabilization of the hydrogen desorption process. The temperature continues to drop with 1 NL/min hydrogen desorption until it becomes difficult to maintain stable hydrogen discharge. The end of this stage is the temperature reduced to the lowest temperature (−6 °C) and the flow rate is not maintained at 1 NL/min, the pressure is reduced to the same as the outlet pressure (0.1 MPa) as well. At this time, the heat transfer performance of the MH tank makes it difficult to supply heat at a flow rate of 1 NL/min hydrogen release. Therefore, this stage is based on maintaining the stability of the hydrogen release by external heat transfer and absorption of the heat released from the cooling down of the alloy.

Stage III: This stage is the low-rate hydrogen release stage. The temperature of the tank gradually rises and the flow rate gradually decreases. The system pressure is always maintained at 0.1 MPa. The hydrogen release process diffuses hydrogen directly into the atmosphere. TC1 and TC2 are basically unchanged at the beginning and maintained at −6 °C, suggesting that the corresponding hydrogen release at the beginning should be very small and gradually released. This stage is the process of hydrogen desorption from the outside alloy to the inside alloy gradually reaching more than 90 %. The only source of heat for hydrogen desorption at this stage is through externally introduced heat. Since the alloy temperature is reduced to its lowest temperature at this stage, it is not possible to provide heat for hydrogen desorption by cooling. Therefore, the rate of hydrogen desorption at this stage is directly related to the efficiency of heat transfer. As the temperature rises, the temperature difference becomes smaller, the heat transfer distance becomes longer, and the alloys in the MH tank that can release large volumes of hydrogen become progressively smaller, and some of the conducted heat needs to be used to warm up the external alloys, making the actual hydrogen releasing alloys receive less heat, leading to a continuous reduction in the flow of hydrogen. Therefore, the alloy can completely release hydrogen at this stage.

Stage IV: The hydrogen desorption reaction is completed and the temperature is returned to room temperature.

The stable hydrogen release from the hydrogen storage tank ends at the end of Stage II, at which time the total hydrogen release is the effective hydrogen release. Therefore, in order to enhance the properties of the tank, the time of Stage II should be kept as long as possible.

3.2.2. Effect of internal heat transfer methods

Fig. 11 shows the effect of different internal structures on the performance of hydrogen discharge at a constant flow of 1 NL/min. With natural convection, H_2 is released at 1 NL/min for 63.5, 84.8, 105.1, and 197.8 min, respectively, from Case 0, Case 1, Case 2, and Case 3. Case 3 shows a performance improvement of 211 % more than Case 0. Chung et al. [71] discovered that adding heat pipes to the tanks with a larger heat transfer area could improve performance by 44 %. The results show that the heat transfer improvement by the internal structure can significantly increase the effective desorption of hydrogen which is released at a rate of 1 NL/min at least. The effective hydrogen release time increased by 23.9 % after replacing ENG with copper foam under natural convection, while the hydrogen absorption time changed little under the same condition as shown in Fig. 8. The performance improvement is more obvious in the hydrogen desorption process than that of the hydrogen absorption process. That's because the flow rate of hydrogen release was controlled during the hydrogen release process. The lower hydrogen release flow rate allowed the tank to absorb heat for a longer time, compared to the higher hydrogen absorption rate, resulting in a 23.9 % performance improvement for Case 2 over Case 1. Case 3 shows the best desorption performance. The effective hydrogen release of Case 3 with the addition of copper fins was 133.2 % more than that of Case 1. This performance improvement is attributed to the increase in heat transfer areas. It is shown that in order to get better heat transfer properties of hydrogen storage tanks, both the effective thermal conductivity and heat transfer area should be increased.

3.2.3. Effect of external heat transfer methods

Fig. 12 shows the effect of different external heat transfer methods on the hydrogen desorption performance of Case 1 at a constant flow of 1 NL/min. It can be seen that the effective H_2 desorbed at 1 NL/min increases with the heat transfer coefficient increasing. The results show that the effective hydrogen release respectively increased by 86.8 % and 102.2 % compared with that under natural convection when the external heat transfer was air cooling and water cooling. This is because air cooling and water cooling can considerably enhance the heat transfer performance of MH beds compared to natural convection. Jana and Muthukumar [80] simulated the effect of different external convective heat transfer coefficient on the MH tanks, and the results showed that the rate of heat transfer improved with an increase in the convective heat transfer coefficient. Therefore, the increase of the external

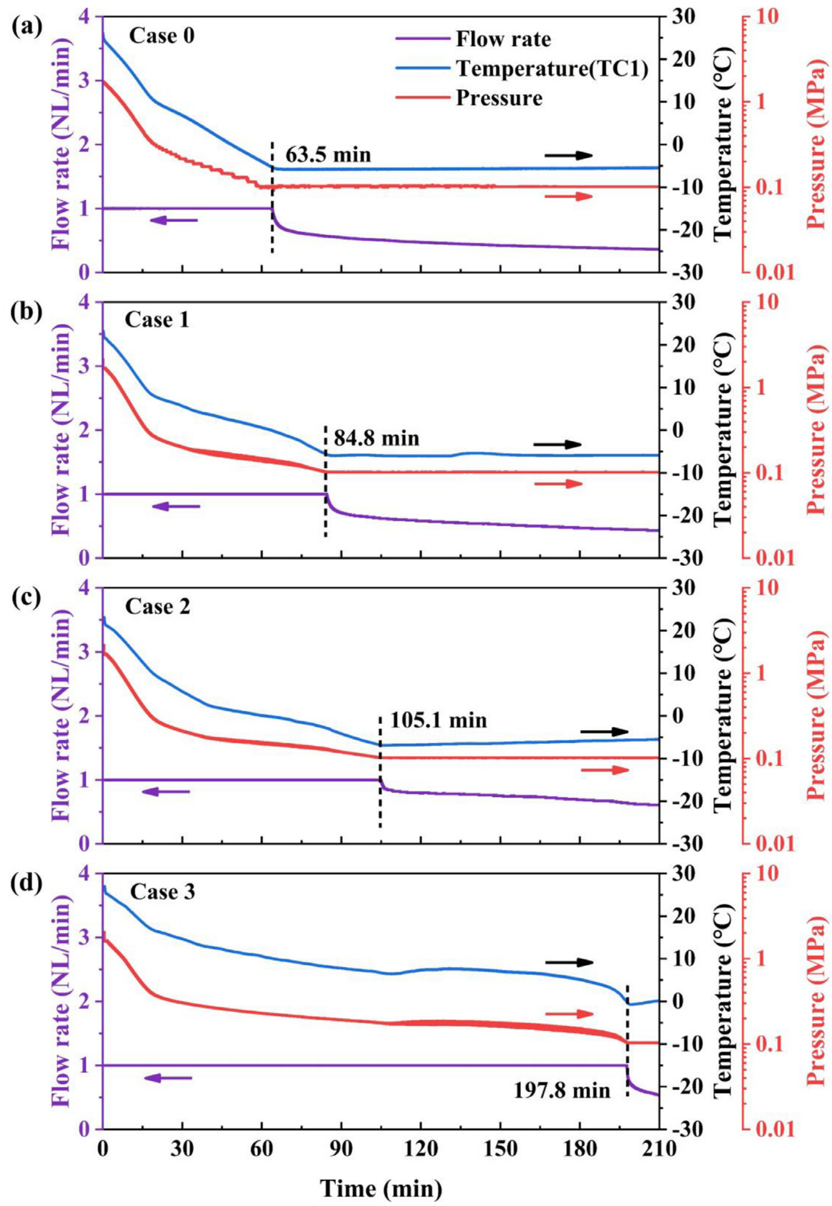

Fig. 11. Effect of different internal structures on the performance of hydrogen desorption at a constant flow of 1 NL/min with natural convection: (a) Case 0; (b) Case 1; (c) Case 2; (d) Case 3.

heat transfer coefficient leads to an improvement in heat transfer of the tank, leading to the remarkable increase of the effective desorbed hydrogen.

For Case 1, the effective desorbed hydrogen only increased by 8.27 % with water cooling compared with air cooling. According to the Van't Hoff equation [29,55,78], the enthalpy of hydrogen desorption for the MH alloy selected in this work is 27.388 kJ/mol. The heat needed for the hydrogen release at 1 NL/min is 20.52 J/s. The estimated convective heat transfer rate is about 20.41 J with air cooling (h = 25 W/(m^2 K) according to Eq. (2). This indicates that the energy supplied in the air-cooling condition is sufficient for the desorption rate of the alloy. However, less than 90 % effective hydrogen was released under air cooling condition for Case 1, which might be caused by the poor heat transfer performance [26] inside the tank, consequently, the heat couldn't be quickly transferred to the core part of the tank. Even though the heat transfer coefficient of water cooling is 10 times more than that of air cooling, the actual hydrogen desorption performance improvement is not significant due to the fact that the heat cannot be efficiently transferred to the core part of the tank due to the poor heat transfer performance of Case 1. It is evident that improving the heat transfer properties inside the tank is necessary for the improvement of effective hydrogen desorption.

3.2.4. Effective hydrogen release

The hydrogen desorption performance of an MH tank cannot be emphasized by the speed of hydrogen release in the same way as hydrogen absorption. The main application of hydrogen is coupling with fuel cells [46]. The hydrogen supply to the fuel cell at any given moment needs to be greater than or equal to the amount of hydrogen it requires. Therefore, the amount of hydrogen that can be steadily released from the tank at the rated flow rate is of concern. Fig. 13 shows the hydrogen desorption time and the effective hydrogen release with different thermal management.

As can be seen from Fig. 13, under natural convection conditions, it can be found that changing the internal structure to improve the internal heat transfer can improve the hydrogen release efficiency to over 90 %. According to a comparison of Case 0 and

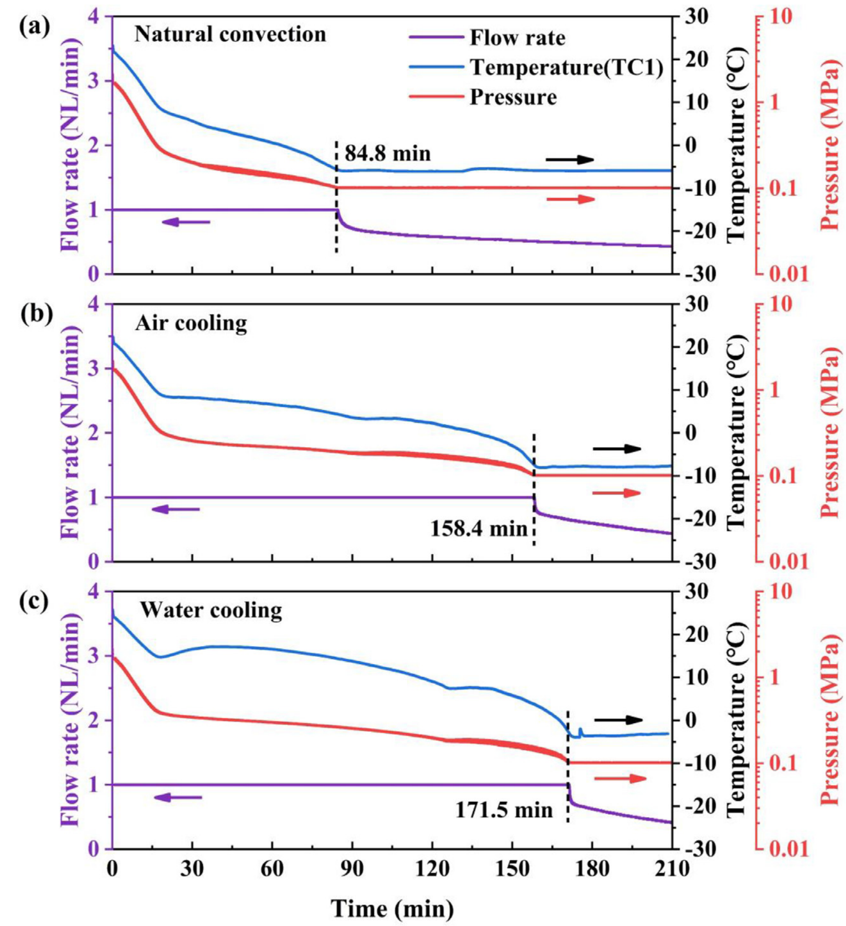

Fig. 12. Effect of different external heat transfer methods on the hydrogen desorption performance of Case 1 at a constant flow of 1 NL/min: (a) Natural convection; (b) Air cooling; (c) Water cooling.

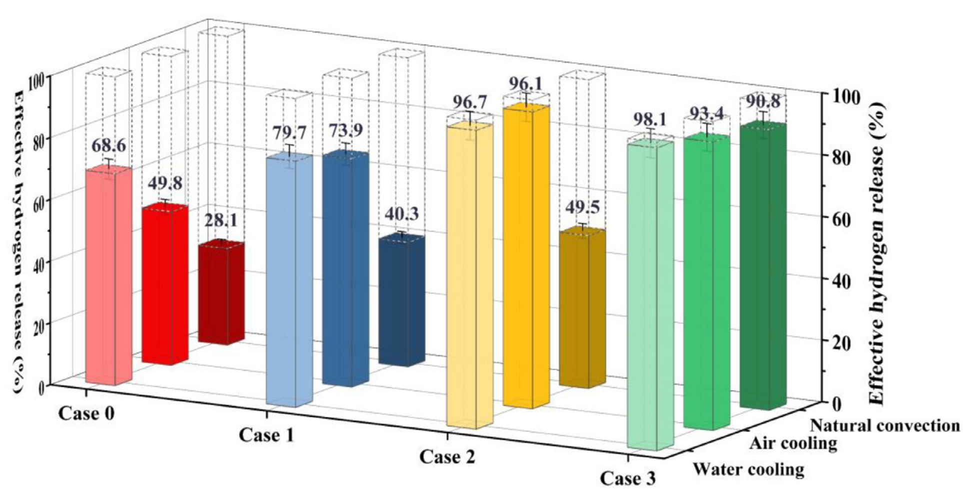

Fig. 13. Effective hydrogen release by using different thermal management methods, with an error bar of 4.8 % resulting from mass flow controller accuracy.

Case 1, the effective hydrogen release of Case 1 is enhanced by 12.2 %, 24.1 %, and 11.1 % under natural convection, air-cooled, and water-cooled, respectively. Among these, the effective hydrogen emission of Case 1 increased by just 11.1 % compared with Case 0 under water cooling condition. This is because the internal heat transfer of Case 1 is still inadequate, hindering the hydrogen release from the core part of the tank, as reported by Chandra et al. [66]. Then, it can be found that for Case 0 and Case 1, even if the external water-cooling method is used, the hydrogen release cannot reach 90 %. When ENG is changed to copper foam which has a better heat transfer property, although the hydrogen release of Case 2 is less than 50 % under natural convection, it can become more than 90 % under air cooling and water cooling. The effective hydrogen release for Case 2 is similar under water cooling and air cooling, indicating that further increasing the external heat transfer coefficient by water cooling has little effect on the improvement of hydrogen desorption when the internal heat transfer property of the tank is above a certain value.

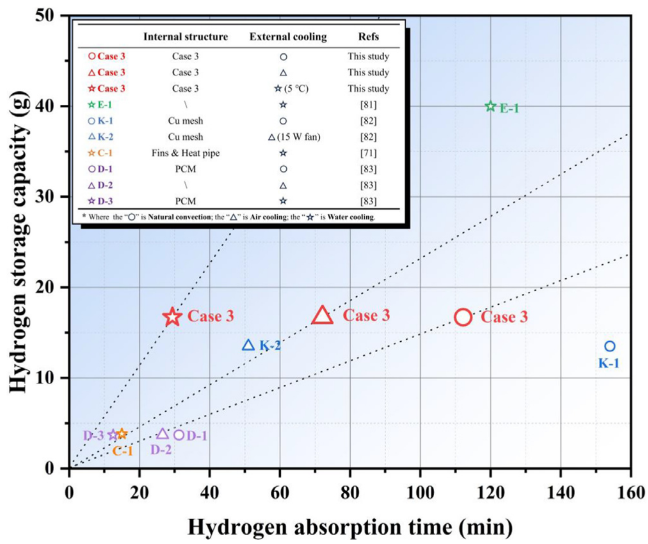

Fig. 14. Comparison of hydrogen storage performance of MH tanks in the present study with the reported studies [71,81–83].

3.3. Effect of thermal management methods on the performance of MH tanks

The results show that as the thermal management performance improves, both the hydrogen absorption and desorption performances of the MH tanks improve. Based on the above results, Case 3 has the best thermal management performance with water cooling, the shortest hydrogen absorption time, and the highest effective hydrogen release. Fig. 14 shows the comparison of the hydrogen storage performance of MH tanks with different heat transfer methods in this work with the reported studies in the last few years [71,81–83]. Elkhatib and Louahlia [81] developed a tank (E-1) for storing 40 g H_2 and the best hydrogen absorption property of the tank was under the cooling water of 5 °C where it took 120 min for the hydrogen absorption. Kim et al. [82] researched the combined effect of copper mesh (K-1) and air cooling with a fan on an MH tank (K-2) filled with 1 kg of La(Ce)Ni_5, and it took 53.1 min for the hydrogen absorption. Chung et al. [71] designed a cylindrical metal hydride tank (C-1) filled with 295 g $LaNi_5$, equipped with a finned sleeve in which a heap pipe was installed, and it took 15 min for the hydrogen absorption under water-cooling conditions. Disli et al. [83] studied the absorption of a metal hydride hydrogen storage tank based on various thermal management strategies by numerical simulation and optimized the design parameters of the phase change material (PCM) and operating conditions. The thermal management strategies of three optimized tanks were PCM wrapped around the MH tank (D-1), external convection on the lateral surface of the tank (D-2), and a hybrid approach including the PCM layer and an internal cooling pipe (D-3). As can be seen in Fig. 14, Case 3 under water cooling condition shows the best hydrogen storage performance. With natural convection, the hydrogen absorption properties of Case 3 are clearly better than those of K-1 which needed a much longer hydrogen absorption time with a smaller hydrogen storage capacity, and D-1 which took less time for hydrogen absorption, but had much lower capacity. With air cooling, the hydrogen absorption efficiency of D-2 is worse than that of Case 3. The hydrogen absorption efficiency of K-2 is slightly better than that of Case 3, because the fan with 15 W was used for K-2 which is higher than that used in this work (2.5 W), and the hydrogen capacity of K-2 is less than that of Case 3. With water cooling, the hydrogen absorption efficiency of Case 3 is much better than those of E-1, D-3, and C-1.

The much higher hydrogen absorption efficiency of Case 3 compared with K-1 under natural convection is due to the fact that only the thermal conductivity of the MH bed for K-1 is improved, whereas not only the effective thermal conductivity through ENG is improved, but also the heat transfer area through fins is improved for Case 3. The reason for the difference between Case 3 and D-2 is that only external thermal management is improved for D-2 while both external and internal thermal management are improved for Case 3. Compared with Case 3, only the heat transfer area increased without increasing the effective thermal conductivity of the metal hydride bed for C-1 and D-3, and there was no internal thermal management for E-1 filled with twice more alloys than Case 3, which are the main reasons for their worse hydrogen absorption efficiency.

Therefore, in order to obtain better performances of MH tanks, it is necessary to combine various thermal management methods to increase the thermal conductivity, heat transfer area, and heat transfer coefficient of the MH tank.

4. Conclusions

This work focuses on the influence of the internal and external heat transfer methods of MH tanks on the performance of hydrogen absorption and desorption. The main conclusions are summarized as follows:

(1) By studying the effect of different external heat transfer methods on the hydrogen absorption and desorption performance of the MH tanks, the performance of the MH tanks can be greatly improved by improving the external heat transfer coefficient. After changing the external environment of Case 1 from natural convection to water cooling, the hydrogen absorption time was improved by 72.8 % and the effective hydrogen release by 97.8 %.
(2) By changing the internal structure of hydrogen storage tanks, it is found that after adding copper fins to the hydrogen storage tank filled with 5 wt.% ENG, the hydrogen absorption time is reduced by 55.6 %, and its stable hydrogen desorption (1 NL/min) has reached 98.1 % of the total hydrogen released under water cooling.
(3) The effective thermal conductivity and heat transfer area of the metal hydride bed play an important role in improving tank performance. However, when the overall effective thermal conductivity of the tank is increased to a certain level, there is little value in further increasing the heat transfer coefficient.
(4) Heat transfer is more important than mass transfer to improve the hydrogen absorption and desorption rate of the hydrogen storage tank. In order to ensure rapid absorption and stable desorption of hydrogen in MH tanks, a suitable internal heat transfer structure can be selected according to the real conditions of use.
(5) In order to obtain better performances of MH tanks, it is necessary to combine various thermal management methods to increase the thermal conductivity, heat transfer area, and heat transfer coefficient of the MH tank.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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